

Course Code	Course Title	L	T	P	C
BACSE201	Models of Computation	3	1	0	4
Pre-requisite	NIL	Syllabus Version			
		1.0			
Course Objectives					
1. To introduce the fundamental concepts of formal languages, grammars, and automata theory to provide a foundation for the theoretical understanding of computation 2. To explore the design and analysis of finite automata, regular expressions, context-free grammar, pushdown automata, and Turing machines, emphasizing their role in language recognition and parsing 3. To analyze problems related to decidability, reducibility, and undecidability, and to understand the limitations of algorithmic computation					
Course Outcomes					
1. Apply various proof techniques such as induction, contradiction, and contraposition in the context of formal languages and automata 2. Design and analyze finite automata and regular expressions to solve lexical analysis problems and recognize regular languages 3. Construct and simplify context-free grammar and apply parsing algorithms to determine syntactic structure in languages 4. Develop pushdown automata and understand parsing strategies such as LL(1) and LR for context-free languages 5. Understand and evaluate the computational power and limitations of Turing machines, and distinguish between decidable, semi-decidable, and undecidable problems					
Module:1	Foundations of Formal Languages and Proof Techniques	5 hours			
Proof techniques: Induction, Contradiction, Contrapositive, Introduction to formal languages and operations - Types of grammars - Chomsky hierarchy- Overview of models of automata					
Module:2	Finite Automata and Regular Languages	9 hours			
Deterministic and Non-Deterministic Finite Automata (DFA & NFA) - Îµ-moves in NFA - NFA to DFA conversion - Minimization of DFA - Regular expressions - Lexical Analysis using FA and Regular Expression, Program Constructs using Regular Expression, Equivalence of finite automata and regular expressions - Regular languages - Pumping Lemma					
Module:3	Context Free Grammars and Languages	9 hours			
Context-Free Grammars (CFG) - Context-Free Languages - Derivations - Ambiguous and Unambiguous grammars - First and Follow - Left Recursion and Left Factoring - Simplification of CFG: Chomsky Normal Form - Greibach Normal Form - CYK Algorithm					

Module:4	Parsing and Pushdown Automata	11 hours
Introduction to parsing: Top-down and Bottom-up - LL (1) Parsing - SLR Parsing - Pushdown Automata (PDA) - Deterministic and Non-deterministic PDA - Interpretation of syntactic statements using PDA.		
Module:5	Turing Machines and Undecidability	9 hours
Turing Machines (acceptors and transducers) - Universal Turing machine and encoding - Church Turing Thesis - Recursive Enumerable Languages -Recursively Enumerable Languages -Non Recursive Languages -Reducibility - Undecidability: Halting Problem - Post Correspondence Problem - Tools: JFLAP, REGEX, LEX, YACC		
Module:6	Contemporary Issues	2 hours
Total Lecture Hours:		45 hours
Tutorial Hours:		15 hours
Text Book(s)		
1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman, " Introduction to Automata Theory, Languages, and Computation ", Pearson Education, 3 rd Edition, 2020		
Reference Books		
1. Peter Linz, Susan H. Rodger, Jones, " An Introduction to Formal Languages and Automata ", Bartlett Publishers, 7 th Edition, 2022 2. Dan A. Simovici, " Introduction to the Theory of Formal Languages ", World Scientific, 2024 3. Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, Sorav Bansal, " Compilers: Principles, Techniques, and Tools, Updated ", Pearson India, 2 nd Edition, 2006		
Indicative Experiments		
Total Laboratory Hours:		0 hours
Mode of Evaluation :Continuous Assessment Test, Digital Assignment, Quiz, Final Assessment Test, Seminar		
Recommended by Board of Studies :		21-05-2025
Approved by Academic Council : No. 78		12-06-2025